

MTD2A_timer

MTD2A: Model Train Detection And Action – Arduino library <https://github.com/MTD2A/MTD2A>

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MTD2A_timer is an easy-to-use, advanced and functional C++ class for non-blocking time control. MTD2A supports parallel processing and asynchronous execution.

The class is among a number of logical building blocks that solve different functions.

Common to all building blocks are:

- They support a wide range of input sensors and output devices
- Are simple to use to build complex solutions with few commands
- They operate non-blocking, process-oriented and state-driven
- Offers extensive control and troubleshooting information
- Thoroughly documented with examples

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Feature Description

MTD2A_timer process consists of 3 functions:

1. `MTD2A_timer object_name ("object_name" , countdownMS);`
2. `object_name.timer ( { RESET_TIMER | START_TIMER | PAUSE_TIMER | STOP_TIMER }, countdownMS);`
Called in `void setup ();`
3. `MTD2A_loop_execute ();` Called as the last instruction in `void loop ();`

The first argument uses the default value and the function can be called with none and up to 2 arguments.

```
MTD2A_timer object_name;  
MTD2A_timer object_name ("Object_name");  
MTD2A_timer object_name ("Object_name", countdownMS);  
Default: ("Object_name", 0 );
```

If `countdownMS = 0`, `object_name.timer ()` ends without further ado and warning 140 is recorded.

If `object_name.timer ( { PAUSE_TIMER | STOP_TIMER }, countdownMS )` is called, `countdownMS` is ignored and warning 142 is recorded.

NOTE: Do not call MTD2A_timer from an interrupt handler (ISR).

Example of event-controlled process

```
// Event if-condition controlled. Recommended for simple solutions
// Read infrared sensor. Short detection: One LED blink.
// continuously detection: continuously LED blink
// Suitable for Unpredictable process flows such as object detection with a sensor
if (IR_sensor.get_processState() == ACTIVE) {
  Serial.println ("Object detected");
  if (timer_GL1.get_processState() == COMPLETE) {
    timer_GL1.timer (START_TIMER, 1000); // 1 second
    green_LED_1.activate (500);          // 0.5 second
  }
}
```

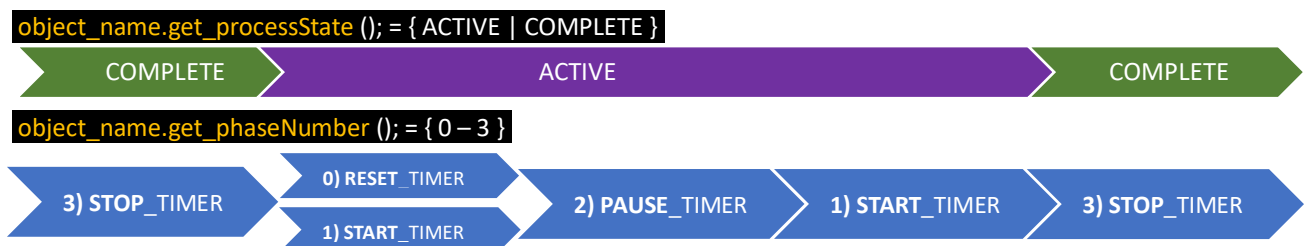
More examples and Youtube demo video:

<https://github.com/MTD2A/MTD2A/tree/main/examples>

DEMO video: https://youtu.be/UU4k4_8GWfM

Process phases

Depending on the current configuration, the process is carried out in between 2 and multiple phases with pauses.



3. The initial phase when the program starts `STOP_TIMER`.
0. When function `object_name.timer (RESET_TIMER);` Is called. Can occur at any time.
1. When function `object_name.timer (START_TIMER);` Is called. May occur several times after each pause.
2. When function `object_name.timer (PAUSE_TIMER);` Is called. May occur several times after start and reset.
3. When function `object_name.timer (STOP_TIMER);` Is called or when the time has expired. `remainTimeMS = 0`.

Global Number Constants:

`RESET_TIMER`, `START_TIMER`, `PAUSE_TIMER` and `STOP_TIMER`

The immediate phase shift can be identified by function: `object_name.get_phaseChange (); = { true | false }`

Process status

When transitioning to `START_TIMER` or `RESET_TIMER`, ProcessState switches to `ACTIVE`.

When transitioning to `STOP_TIMER`, the processState switches to `COMPLETE`.

ProcessState remains `ACTIVE` during `PAUSE_TIMER`.

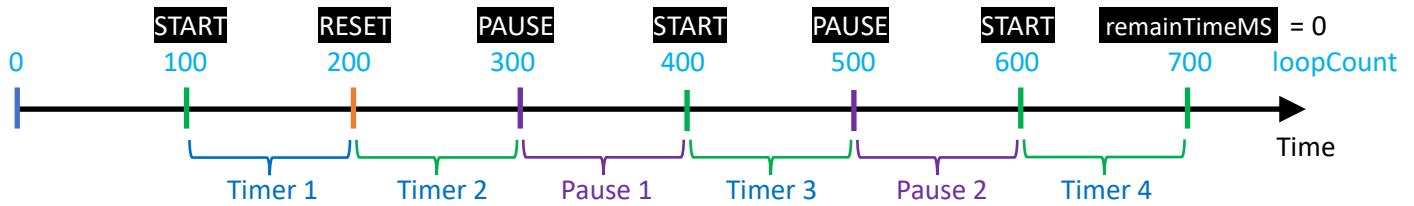
Timing

The time period can be set when the object is instantiated. Subsequently, new time periods can be defined with the function: `object_name.set_countDownMS ( { 0 - MAX_TIME_MS } );` But only when processState is `COMPLETE`

Refer to the document MTD2A.PDF and the section "Cadence", "Synchronization" and "Execution speed".

Example of time-controlled process

Timer function is activated at 1000 millisecond intervals (100 loopCount).



Timer 1 deleted

ElapsedTimeMS = Timer 2 + Timer 3 + Timer 4 = approx 3.000 milliseconds.

remainTimeMS = countdownMS – elapsedTimeMS = 0 milliseconds.

PauseTimeMS = Pause 1 + Pause 2 = approx 2.000 milliseconds.

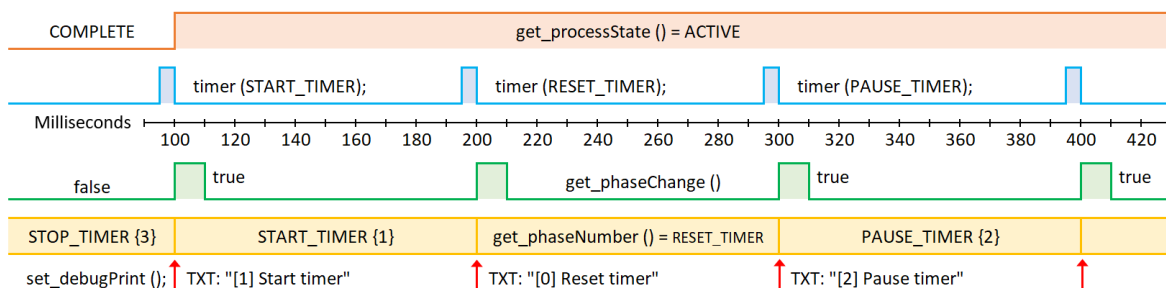
MTD2A_timer T1 ("T1", 3000);

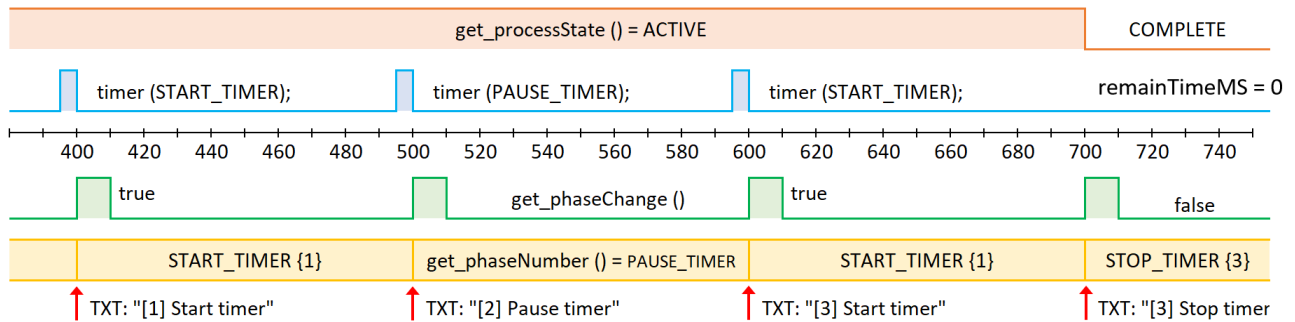
```
// Standard Loop executing time = 10 milliseconds
if (loopCount == 0) { T1.set_debugPrint (); }
if (loopCount == 100) { T1.timer (START_TIMER); } // 1 second
if (loopCount == 200) { T1.timer (RESET_TIMER); } // 2 seconds
if (loopCount == 300) { T1.timer (PAUSE_TIMER); } // 3 seconds
if (loopCount == 400) { T1.timer (START_TIMER); } // 4 seconds
if (loopCount == 500) { T1.timer (PAUSE_TIMER); } // 5 seconds
if (loopCount == 600) { T1.timer (START_TIMER); } // 6 seconds

loopCount++;
if (T1) { // equivalent to: phaseChange == true && phaseNumber == STOP_TIMER;
    T1.print_conf ();
    loopCount = 0;
}
```

IDE Serial Monitor

```
36:04 -> T1 [1] Start timer
36:05 -> T1 [0] Reset timer
36:06 -> T1 [2] Pause timer
36:07 -> T1 [1] Start timer
36:08 -> T1 [2] Pause timer
36:09 -> T1 [1] Start timer
36:10 -> T1 [3] Stop timer
```





Pause time limit

The total time budget for one timer run is limited: **countDownMS + accumulated pause time must not exceed MAX_TIME_MS** (4,200,000 milliseconds $\approx$ 70 minutes). The reason is that all time calculation uses 32-bit microsecond arithmetic, which wraps around after approx. 71.6 minutes. The limit leaves a safety margin below the wrap-around point.

The limit is monitored automatically every loop while the timer is paused. If the budget is exceeded, the timer does not deliver wrong results — instead it is stopped safely and warning **141** is issued: *"Time + pause exceeds MAX_TIME_MS"*

Practical advice: **PAUSE_TIMER** is intended for pauses of limited duration. If a process must wait for a long or unpredictable time (for example a sensor event that may take hours), do not leave the timer paused. Use **STOP_TIMER** instead, and **START_TIMER** / **RESET_TIMER** it again when the process resumes.

Print_conf();

```
object_name.print_conf();
```

Maximum time accuracy with Arduino ESP32 and cadence **DELAY_10MS**

```
MTD2A_timer:
-----
objectName      : T1
processState    : COMPLETE
phaseText       : [3] Stop timer
debugPrint      : ENABLE
globalDebugPr   : DISABLE
errorPrint      : DISABLE
globalErrorPr   : ENABLE
errorNumber     : 0 OK
countDownMS     : 3000
elapsedTimeMS   : 3000
remainTimeMS    : 0
startTimeMS     : 4144
stopTimeMS      : 9144
pauseTimeMS     : 2000
pauseBeginMS    : 7144
pauseEndMS      : 8144
```

If there are multiple breaks during the time period, it shows the total pause time **pauseTimeMS**, but **pauseBeginMS** and **pauseEndMS** only appear for the last pause.

```
MTD2A_print_conf();
```

```
MTD2A_base:
-----
globalDebugPrint : DISABLE
globalErrorPrint : ENABLE
globalObjectCount: 1
globalDelayTimeMS: 10
globalSyncTimeMS : 8766
lastElapsedTimeMS: 0.00
maxElapsedTimeMS : 0.00
timeOverrunCount : 0
totalLoopCount   : 701
```

Set and Get Features Overview

Set functions	Comment
set_countDownMS ({0 - MAX_TIME_MS});	Set new count down time in milliseconds
set_debugPrint ({ENABLE DISABLE});	Activate print phase number and text
set_errorPrint ({ENABLE DISABLE});	Activate error messages

Get functions	Comment
get_processState (); return bool {ACTIVE COMPLETE}	Process state
get_phaseChange (); return bool {true false}	Momentarily phase change (one loop time)
get_phaseNumber (); return uint8_t {0 - 3}	RESET_TIMER = 0, START_TIMER =1, PAUSE_TIMER = 2, STOP_TIMER = 3
get_startTimeMS (); return uint32_t milliseconds	Time set at start/reset
get_stopTimeMS (); return uint32_t milliseconds	Time set at stop
get_pauseTimeMS (); return uint32_t milliseconds	Accumulated pause time (sum of all pause periods). Zero if no pause initiated
get_remaingTimeMS (); return uint32_t milliseconds	Remaining time since start/reset
get_elapsedTimeMS (); return uint32_t milliseconds	elapsed time since start/reset
get_reset_error (); return uint8_t {0-255}	Get error/warning number and reset Error [1 - 127] warning [128 - 255]

Operator overloading	Function
object_name & !objectName	bool phaseChange == true && phaseNumber == STOP_TIMER
object_name_1 == object_name_2	bool processState_1 == processState_2
object_name_1 != object_name_2	bool processState_1 != processState_2
object_name_1 > object_name_2	bool processState_1 = ACTIVE & processState_2 = COMPLETE
object_name_1 < object_name_2	bool processState_1 = COMPLETE & processState_2 = ACTIVE